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RESEARCH ARTICLE

Data-Driven Analysis and Modeling for Energy Storage and PV Integration in Trolleybus Networks

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ABSTRACT The electrification of public transportation systems plays a crucial role in achieving sustainable urban mobility. While trams and trolleybuses have demonstrated energy-efficient operation for a long time, their integration with renewable energy sources and energy storage systems (ESS) remains an important area of research. This paper presents a data-driven methodology for analyzing trolleybus energy flows and determining optimal ESS and photovoltaic (PV) deployment strategies using real-world operational data from the City of Riga. A comprehensive dataset was collected from on-board loggers installed in trolleybuses operating across four major routes. A large dataset containing key traction values was upsampled to enable more detailed analysis. Heat-map visualization techniques were used to identify locations of high energy exchange, revealing candidate zones for ESS placement within the DC overhead contact system. Complementary PV production data collected over eight years from a reference rooftop system were statistically analyzed and then adjusted to represent a standardized 1 kW output for scaling purposes. MATLAB Simulink models were developed to simulate substation behavior under varying load and regeneration conditions. The results indicate that integration of supercapacitors can significantly stabilize line voltage and improve recuperation efficiency. ESS further choice analysis was performed using realistic solar yielded energy and load data under different operational strategies. Combining large-scale operational datasets with simulation modeling provides a powerful framework for optimizing renewable integration and energy storage in public transport grids, paving the way toward more efficient and resilient electric mobility infrastructure.

INDEX TERMS Public transportation, photovoltaic systems, energy storage, functional data analysis, power grids.

1. INTRODUCTION

The deployment of electric vehicles into modern society is progressing rapidly, particularly in the field of individual transportation. At the same time, electric transport has been a part of public mobility systems for a long time. Trams and trolleybuses have proven to be among the most

efficient solutions for achieving sustainable urban mobility and reducing environmental impact. Electricity generation in centralized power plants, combined with efficient energy utilization through public transportation, minimizes losses primarily to grid transmission. This approach not only alleviates urban traffic congestion but also provides higher energy efficiency per passenger in comparison to private transport. Consequently, although these systems were introduced long ago, trolleybuses and trams still play a significant role in

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modern sustainable city concepts. One of the simplest ways to enhance overall system efficiency is to utilize recuperative energy connecting the transport system units into a common energy-sharing system [1], [2].

Recent advancements in Energy Storage Systems (ESS) [3], [4], [5], [6], [7], [8], [9] and Renewable Energy Sources (RES) have further increased overall efficiency and reduced the load on the AC grid [6], [10], [11], [12], [13], [14], [15], [16], [17]. These technologies contribute to a more stable electricity consumption by smoothing demand peaks. In particular, the solar generation profile closely aligns with daytime traffic patterns in public transport, positioning solar energy as one of the most promising sources for grid load reduction in urban transport systems. In recent years, several studies have emphasized the importance of integrating energy storage technologies into modern power systems to support the transition toward low-carbon and flexible energy infrastructures. For example, the role of various storage technologies and alternative fuels in the transformation of power systems toward climate-neutral operation was analyzed in [18]. Furthermore, the optimal combination of renewable energy sources and storage solutions has been investigated for the Baltic power system, demonstrating that properly deployed storage technologies can significantly improve system flexibility and reduce transition costs [19].

An interesting concept is proposed in [20], providing energy flow analysis of railway power supply system considering circulation currents and train on-board resistances. In contrast, this study focuses on a full utilization of regenerative energy while smoothing the load profile.

The motivation for this research stems from the observation that a large number of studies focus primarily on theoretical data or isolated short-term examples [21] and [22] using optimization approaches. Many works propose aligning transport timetables to synchronize braking and acceleration cycles of trams, trolleybuses, or metro trains to balance the energy flow [1], [23], [24]. While this theoretically sounds promising, practical implementation could face significant challenges. Passenger safety and comfort requirements, fluctuating passenger flows, traffic congestion, and rush-hour delays in metro operations result in substantial deviations from idealized conditions. Hence, operating a large urban transport system under perfectly synchronized conditions is nearly impossible and difficult to be maintained in practice.

Onboard ESS is a notable concept overviewed in [25] and [26], with the potential to reduce copper losses compared with substations equipped by ESS. However, the integration of onboard ESS will result in additional weight, which, in turn, will result in energy consumption increase as the extra weight increases both dynamic and static mode motor torque, increases braking path and reduces the useful volume. The paper proposes a trade-off, to find optimal locations for installation of smaller ESSs and connect them to Overhead Contact System (OCS). Sometimes ESS equipped substations are the best solution, for example, for the trains like in [27].

Therefore, research based on real-world operational data becomes essential. Selecting optimal locations for ESS installations and photovoltaic (PV) systems according to actual transport demand patterns can lead to a more sustainable and adaptable infrastructure. Such a system can dynamically accommodate timetable changes and daily operational variations due to unforeseen delays or route adjustments.

In the City of Riga, almost every trolleybus is equipped with a data logger device. Consequently, trolleybuses represent one of the most modernized elements within the city's electric transport fleet. Some trams, in contrast, are less technologically advanced. For this reason, the trolleybus network was selected as the primary focus of this study. However, in certain cases, the same OCS substation supplies both trams and trolleybus networks. This introduces some discrepancies when modeling the voltage and current dynamics over time. However, the general objective—analyzing the impact of ESS and PV integration on the total grid demand—remains valid.

The onboard loggers collect a wide range of operational parameters, including vehicle velocity, grid voltage, current, motor torque, and various control signals such as braking and door sensors, daylight sensors, and driving modes. The public transport operator in Riga, *Rīgas Satiksme*, manages a fleet of 346 trolleybuses. Given the one-year project duration and the limited scope of voluntary cooperation from *Rīgas Satiksme*, four of the most frequently used routes (out of 22 total) were selected for analysis. These routes collectively cover approximately half of the overall trolleybus network area and constitute the basis of the analysis presented in this paper.

Public electric transportation represents a significant source of energy consumption within modern urban environments. However, the regenerative (recuperative) energy produced during braking is not fully utilized, leading to inefficiencies in the overall system performance. At the same time, the total traffic flow — and, consequently, the total energy demand — closely follows the solar day profile. This correlation highlights the potential of integrating RES, particularly photovoltaic (PV) systems, into public transport infrastructure. The implementation of such energy-saving and renewable solutions can substantially reduce electricity consumption from the grid. To achieve this effectively, it is necessary to thoroughly analyze the existing infrastructure, define general sizing guidelines, and assess how the proposed measures would influence the system's overall performance and stability—an aspect that represents a novel contribution of this paper.

II. METHODOLOGY

A. DATA MINING AND MAKING A COMMON DATASET FOR TROLLEYBUSES

The primary dataset collected for this study contains more than six million rows and seventy-three columns of logged values including voltage, current, motor torque, door opening status, diesel engine status, and other operational parameters. The time interval between most recorded data points

is approximately 0.1 s; however, since the logging device operates based on a 100 m distance reference, the time intervals increase as the trolleybus velocity decreases. The logger records raw GPS coordinates at intervals of approximately 100 – 300 m. These coordinates are unconditioned and may deviate from the actual route by approximately 5 m with deviations reaching up to 15 m off the real road location in some cases. Consequently, manual corrections were applied to the data. The logging device is a proprietary closed box unit provided by Skoda. The manual does not contain specific information on data acquisition filtering and provides a description of the mentioned above acquisition algorithm only. However, based on several data graphs it can be concluded that the ADC values of voltages and currents have harder filtering as the values have smoother variations than, for example, torque value series. No additional filtering was applied during data processing to preserve the original data integrity. After acquisition of raw GPS data, missing GPS coordinates in high-rate samples were reconstructed via interpolation using odometer readings and path reconstruction from neighboring GPS-tagged data logger records, supported by mapping microservices. This represents one of the potential sources of inaccuracy in the route visualization. Nevertheless, the offset effect is minor: errors of approximately 5 m between 100 m sampling points are negligible compared to the general GPS drift and do not significantly affect map-based route reconstruction using mapping and routing microservices. To provide spatial information for each logged sample at a temporal resolution of 0.1–0.3 s, data were interpolated using odometer-based distance measurements. GPS-tagged points recorded at intervals of 100–250 m were used as reference anchors. The positions of intermediate samples were then reconstructed using map-microservice queries in combination with cumulative trip-distance data retrieved from the database.

The dataset corresponding to the 6th of January 2025 was provided by *Rīgas Satiksme*. Despite being recorded on a winter date, the weather conditions were more typical of spring or autumn, with ambient temperatures of around +12 °C and no snow cover. These conditions are representative of the most common operating environment in Latvia; therefore, the obtained results can be considered indicative of typical annual conditions, including standard Heating, Ventilation and Air Conditioning (HVAC) operation and passenger load patterns. Data collection resulted in a dataset of approximately 6.2 million rows by 73 columns of unique measurements for a single 24-hour period. After extracting unique original entries, the dataset was upsampled using interpolation to achieve a uniform time resolution of 0.1 s, thus ensuring consistency and facilitating further data processing. The data was interpolated in respect to time, and the results were further used only for MATLAB simulation that can be seen in further chapters.

B. DATA MERGING AND PROCESSING

All available datasets were combined into a single unified dataset once the data collection process was completed. The

Python Pandas library was selected as the primary data processing framework, as it is one of the most widely used tools for relational data handling. Moreover, since Pandas library is built on top of NumPy, it provides efficient performance for numerical operations and supports multithreaded computation without requiring additional external libraries or custom parallelization.

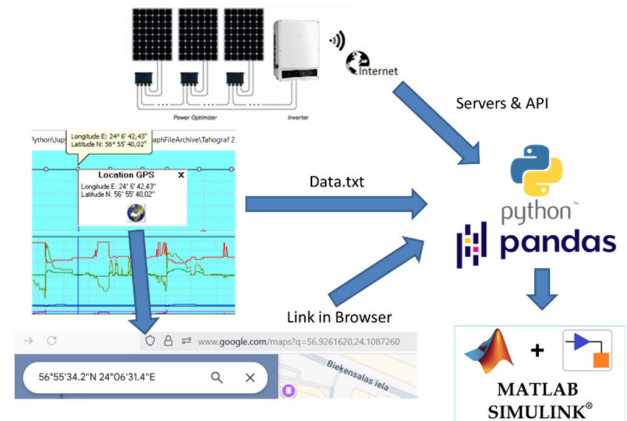


FIGURE 1. Workflow diagram for data collection and processing.

Fig. 1 illustrates the workflow used for collecting and processing data to achieve the objectives of the research. One of the major challenges encountered was related to GPS data acquisition, as this information is crucial for identifying typical city locations influencing energy flow patterns. The original Tachograph software used for trolleybus loggers exports most operational data—such as torque, velocity, and Boolean values corresponding to various trolleybus modes and equipment states—directly. However, GPS coordinates must be obtained through a separate function that accesses Google Maps services through a dedicated link. Since the original *.tgf* files are encoded, this is the only available method for retrieving GPS coordinates.

After extraction, GPS coordinates were time- and distance-matched with the main dataset and further interpolated to achieve a 0.1 s temporal resolution for all records. With the unified sampling rate established, the resulting dataset can be easily merged with additional datasets.

For improved grid modeling, the grid resistance was calculated according to the Thévenin theorem, expressed as:

$$\begin{cases} E = V_1 + RI_1 \\ E = V_2 + RI_2 \end{cases} \quad (1)$$

From (1), the resistance value R can be derived as:

$$R = \frac{V_2 - V_1}{I_1 - I_2} \quad (2)$$

It is important to note that this approach is applicable only to consecutive database rows with differing current values; otherwise, division by zero occurs.

To address missing values (NaN), the statistical mode (most common value) method was applied. Once the grid

resistance was computed, it enabled estimation of the equivalent rectifier voltage, which was compared with real substation voltage measurements. Although substation data were recorded at a much lower sampling rate, the comparison confirmed that the calculated voltages closely correlated with actual DC source values.

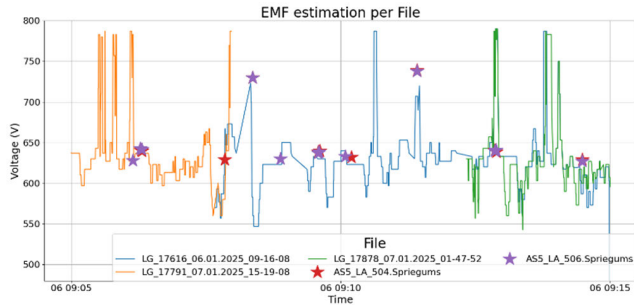


FIGURE 2. Estimated EMF voltage values by processing on-board logger data (solid), LG_17xxx stands for unique logging file name for each trolleybus and actual substation measurements (stars); AS5_LA_50x stands for the name of substation section.

As shown in Fig. 2, the measured and estimated electromotive force (EMF) voltages do not coincide perfectly but demonstrate strong relative agreement, validating the proposed estimation approach.

C. POWER ESTIMATION FOR RETRIEVING DISCHARGED BRAKING POWER

In contrast to internal combustion vehicles, trolleybuses operate without multi-speed transmissions; therefore, traction motor torque, velocity, and mechanical output power maintain a fixed functional relationship, assuming the electrical machine's efficiency curve remains approximately constant. In this study, we adopt this simplification and use the available instantaneous torque and vehicle velocity measurements to estimate mechanical power.

Vehicle velocity is first converted to wheel angular speed using the known wheel diameter. The motor shaft speed is then inferred through an approximate gear ratio, calculated by matching the maximum rated speed of the trolleybus and motor maximum rated RPM from manufacturer data. After normalization of all variables to SI units a linear coefficient relating vehicle velocity to motor angular velocity can be obtained. Although this approach introduces uncertainty, it provides an analytically convenient proportionality factor that can be refined later — for example, using a dedicated linear regression model.

To obtain reliable regression data, only samples with positive mechanical power and electrical consumption above 20 kW are retained, thereby excluding low-speed or idle periods where auxiliary loads or intermittent logging dominate. These filtered mechanical power estimates are then paired with measured grid power (computed from line voltage and current) to identify the regression coefficients. The resulting

model is subsequently applied to the full dataset.

$$X = \begin{bmatrix} T(1) & v(1) & U_{DC}(1) & I_{DC}(1) \\ T(2) & v(2) & U_{DC}(2) & I_{DC}(2) \\ \vdots & \vdots & \vdots & \vdots \\ T(N) & v(N) & U_{DC}(N) & I_{DC}(N) \end{bmatrix} \quad (3)$$

where $T, v, U_{DC}, I_{DC} \in \mathbb{R}^N$ represent torque, velocity, voltage and current respectively.

The estimated motor angular velocity and mechanical power are computed as:

$$\omega_m^{(EST)} = k_v v \quad (4)$$

$$P_{mech}^{(EST)} = T \odot \omega_m^{(EST)} = k_v (T \odot v) \quad (5)$$

and the grid-side power is given as:

$$P_{grid} = U_{DC} \odot I_{DC} \quad (6)$$

where $\odot$ denotes the elementwise (Hadamard) product

Filtering mask:

$$m = \begin{cases} 1, & P_{mech}^{(EST)}(i) > 20\,000 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

The filtered sets are:

$$\begin{aligned} P_{grid_sel} &= \{P_{grid}(i) | m_i = 1\} \\ P_{mech_sel} &= \{P_{mech}^{(EST)}(i) | m_i = 1\} \end{aligned} \quad (8)$$

The regression model is assumed as:

$$P_{grid_sel} = \alpha P_{mech_sel} + \beta + \varepsilon \quad (9)$$

which can be expressed in matrix form as:

$$P_{grid_sel} = A\theta + \varepsilon \quad (10)$$

where

$$\begin{aligned} A &= [P_{mech_sel} \quad 1] \\ \theta &= \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \end{aligned} \quad (11)$$

The closed-form least-squares solution of regression parameter vector θ is:

$$\theta = (A^T A)^{-1} A^T y \quad (12)$$

That results in

$$P_{grid_pot} = \alpha P_{mech}^{(EST)} + \beta \quad (13)$$

Since the goal is to construct a dataset that replicates the actual grid consumption while retaining the full information on regenerative braking, the corrected power vector is defined as:

$$P_{grid_pot_corr} = \begin{cases} P_{grid}(i) & , P_{grid}(i) > 0 \\ \alpha P_{mech}^{(EST)}(i) + \beta & , \text{otherwise} \end{cases} \quad (14)$$

The shapes of the obtained datasets are compared in Fig. 3. As shown in Fig. 3, the grid power estimated from torque and velocity exhibits more rapid variations than those

obtained directly from voltage and current measurements, which appear to be more heavily filtered by the onboard logging system. Despite these differences, the regression achieves an R^2 score of $96.0 \pm 1.6\%$ for validation values and $95.8 \pm 2.1\%$ for 38 out of 40 trolleybuses, while the remaining two trolleybuses exhibited much lower values. These two logging files appear to be corrupted, or the associated logging closed boxes may have malfunctioned. A symbolic search for non-linear function to approximate analytic function with genetic algorithm resulted in 2-4% worse results than those presented above. Visual inspection of calculated power profiles obtained using more sophisticated approximation also indicated inferior performance. In general, this confirms a strong correlation between estimated and measured data. To eliminate idle biasing, data intervals with time updates exceeding 0.3 s were excluded.

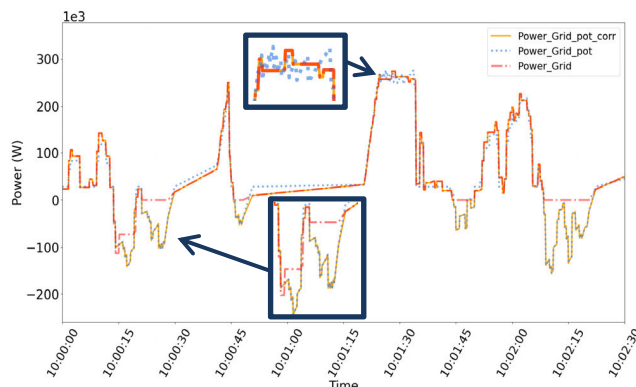


FIGURE 3. Comparison of measured grid power: calculated (dash-dotted red line), estimated from torque and speed sensor (dotted blue line), corrected vector that takes consumption from calculated and braking energy from estimated (solid orange).

When this approximation is applied to the complete dataset encompassing all trolleybuses, the analysis indicates that the current utilization of regenerated energy is approximately 45%, corresponding to 6 % of total grid energy consumption. If an ESS were integrated into the grid, allowing full utilization of regenerative energy, the total grid consumption could be reduced by an additional 8 %, demonstrating a significant potential for efficiency improvement. Comparable saved energy calculations were reported in [5] and [27]. If total energy consumption of trolleybuses during a year is considered, which is 25.68 GWh (data for year 2025.), it is 2.05 GWh or, if the average price is assumed to be 0.11 EUR/kWh, 226'034 EUR. This distribution is shown in Fig. 4.

D. DATA FOR SOLAR ENERGY ESTIMATION

The rooftop of Riga Technical University (RTU) Faculty of Computer Science, Information Technology and Energy is equipped with PV panels. The grid-tied inverter continuously logs the generated energy, providing a comprehensive dataset spanning eight years. This dataset enables extrapolation of typical energy yield patterns under representative weather

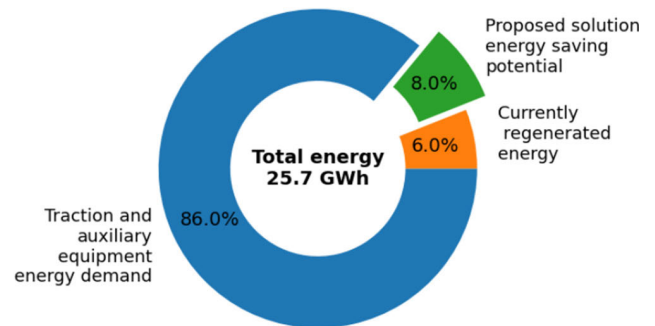


FIGURE 4. Annual energy distribution in the trolleybus system: Current consumption and regeneration versus energy saving potential.

conditions. The rooftop-mounted configuration ensures minimal partial shading, thus providing accurate reference data for expected PV performance.

The policy of solar panel integration into the transport power supply network can follow three main strategies [28]:

- Direct utilization: All generated solar energy is consumed as it is produced.
- Accumulated supply: Solar energy is stored and fed into the grid based on demand.
- Grid-connected contribution: The PV system directly supplies energy to the city grid, reducing substation demand.

The first approach is limited by physical installation size. To design such systems, the maximum expected daily energy generation must be taken as a reference for sizing. Its main advantages are the absence of auxiliary equipment and the shortest payback period, as the reduction in grid consumption directly translates to energy savings. The drawback is that energy reduction occurs only during daytime hours, while demand remains high during evening and morning peak periods when passenger transport activity is still significant.

The second strategy is more expensive but aligns closer with the sustainable development goals of modern cities. Although the environmental impact and resource limitations of lithium-based storage systems are under discussion, large-scale PV plants combined with energy storage can significantly reduce grid dependence. During summer months, some substations may even operate nearly independently from external power sources.

For quantitative evaluation, a statistical model of solar energy generation was developed based on the RTU rooftop dataset. The data covers an eight-year period, and the analysis includes calculation of percentile-based performance indicators—specifically, the 10th percentile (p10) representing worst-case conditions, the 90th percentile (p90) representing best-case performance, and mean and median values for each day of the year. The PV yield data were recorded with a 15-minute sampling interval. Real measured data and data analysis values are represented in Fig. 5 and Fig. 6.

Normalizing the solar yield data to a 1 kW reference enables size-independent reference suitable for system dimensioning. This normalization facilitates the estimation of photovoltaic system output for any installation area or rated power capacity within the analyzed region.

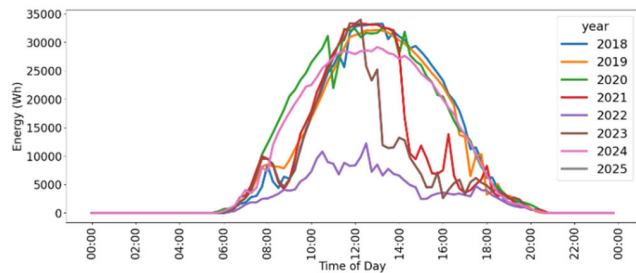


FIGURE 5. PV solar generation data for eight years on 5 january.

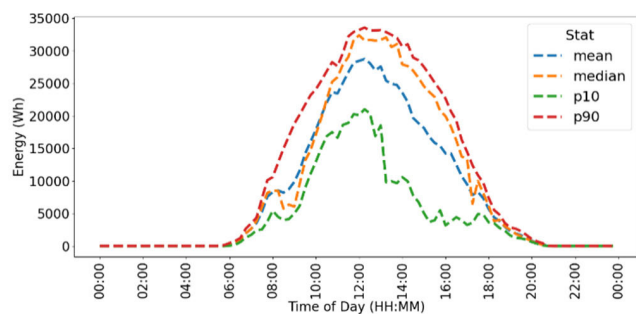


FIGURE 6. Expected PV yield on 5 january. Statistical data: algebraic mean, median, procentile 90 best, procentile 10 worst.

III. DATA PROCESSING AND ANALYSIS

A. MAP VIZUALIZATION

A convenient and intuitive way to identify and visualize the most critical areas and geographical points of interest is through heat maps and coordinate tagging on digital maps. These visualization techniques help highlight regions of high energy exchange activity. For example, the optimal locations for ESS are typically found near areas characterized by high energy consumption and frequent braking events. In systems not equipped with ESS, such locations are often represented by regions on the map with increased voltage levels recorded in the dataset.

The underlying concept is that a high voltage level indicates insufficient trolleybus traffic density to absorb the regenerative (recuperative) energy. Consequently, the DC-link voltage rises until the excess energy is dissipated through the braking chopper and resistors, rather than being reabsorbed by other nearby vehicles. Several reviewed studies present idealized models assuming perfectly synchronized public transport schedules; however, such conditions are rarely achievable in real-life urban operations.

Fig. 7 shows the analyzed dataset column from torque and velocity parameters, normalized against key indicators

of grid power consumption, including both peak and average values. The resulting negative power or current flow values reflect realistic estimates of potential recuperative energy. This approach is necessary because the directly measured grid power—computed from voltage and current—is limited by the voltage-controlled chopper, diverting the excess energy into the braking resistors. Therefore, the power estimation algorithm based on trolleybus velocity and motor torque plays a crucial role in an accurate assessment of the available recuperation potential.

However, determining the best ESS installation sites also requires accounting for grid peak consumption regions. Since the most beneficial ESS locations are where recuperation and consumption coincide spatially, a composite heat map was generated (Fig. 8) that merges both datasets into one.

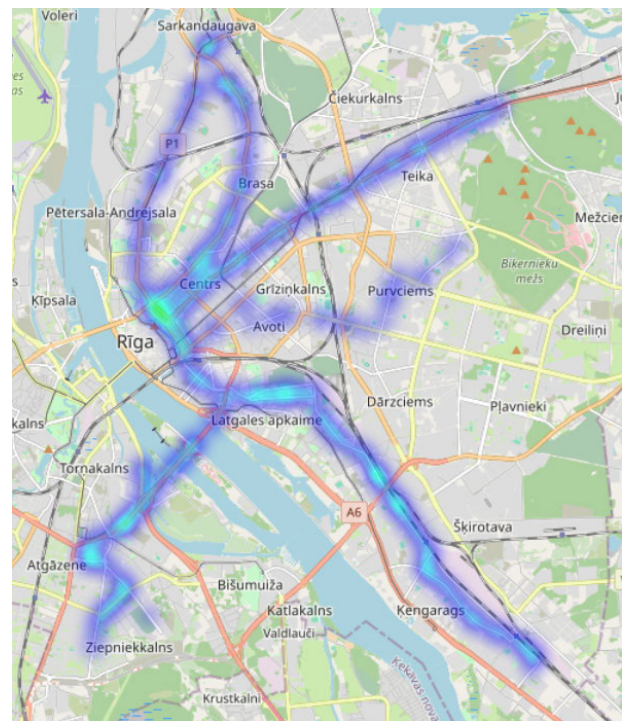


FIGURE 7. Spatial distribution map of voltage higher than 720 V and negative potential power greater than 100 kW available for grid return.

The map in Fig. 7 illustrates a specific area containing potential ESS installation candidates. It should be noted that these points are not intended as fixed locations. Each circle represents an approximate 200 m spatial window, calculated using a 15-second rolling average over time. The heat map visualizes both daytime and nighttime energy consumption and regeneration behavior. The identified candidate points serve as guidelines for minimizing circulation of currents within each substation boundary.

The locations may be adjusted within single substation zones, ideally aligning with heat map regions showing both strong consumption and significant recuperation potential. Furthermore, close proximity of the circles of different color indicators—such as green (regeneration prevailing) adjacent

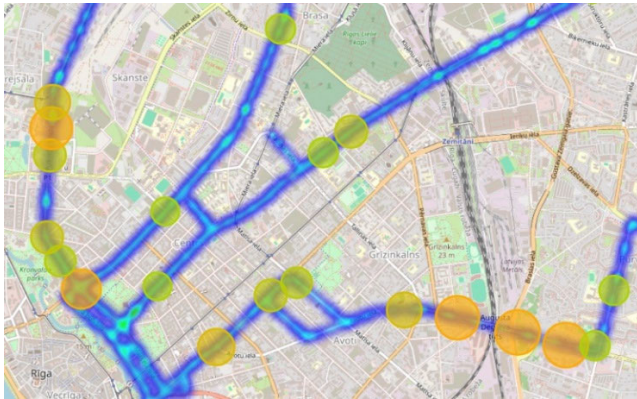


FIGURE 8. Spatial distribution map of combined recuperation and consumption data with OCS line ESS location candidates: orange circles indicate consumption dominance, and green circles denote regeneration dominance. Circle size corresponds to the absolute power magnitude.

to yellow (balanced) or orange (consumption prevailing)—suggests regions where braking and acceleration phases overlap along the route. These overlap zones are particularly promising for ESS installation, as they enable the most effective energy exchange between vehicles.

B. SPOTTING LOCATIONS FOR PROSPECTIVE PV ENERGY PLANTS AND SYSTEM SIZING

The dense development of urban areas significantly limits the available space for large-scale photovoltaic (PV) system installations. Nevertheless, several suitable locations can still be identified. One of the most promising options includes trolleybus and tram depots, which often have extensive open areas that can be utilized for solar energy generation. Research suggested upgrading these depots with roof or semi-open shelter structures, designed both to protect vehicles from rain and snow during winter, late autumn and early spring and to serve as elevated mounting platforms for PV panels. Such dual-purpose designs would simultaneously improve operational reliability and contribute to renewable energy generation.

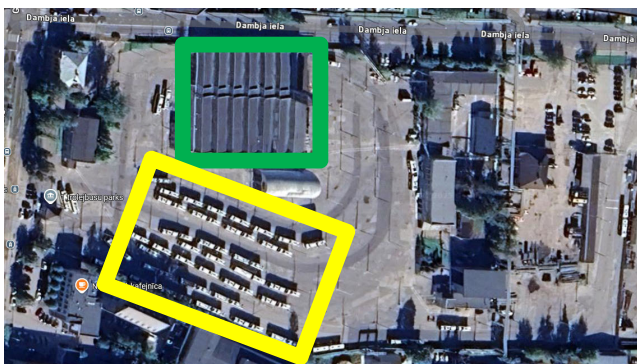


FIGURE 9. Trolleybus depot No. 1 (located in Riga). Rooftop area (green) suitable for PV installation and adjacent parking area (yellow) potentially covered with PV-equipped shelters.

As shown in Fig. 9, the trolleybus depot still contains a considerable unused rooftop and open-space area that can be employed to enhance solar energy yield. Covering these parking areas with gable-roof shelters, as illustrated in Fig. 10, represents an effective approach.

When properly oriented, gable roofs can help to distribute the PV generation curve, reducing peak power while extending the duration of production throughout the day. This is a well-established solution for PV systems that operate without energy storage. Furthermore, such roofs offer additional practical benefits: protection from precipitation, mitigation of corrosion processes, and prolonged vehicle body lifespan, both reducing maintenance costs and increasing overall fleet longevity.



FIGURE 10. Proposed gable-roof trolleybus shelters intended as a base for photovoltaic panel installation.

In addition to depots, several other unused or under-utilized urban areas could potentially host PV systems. However, such sites require individual assessment, as legal and zoning constraints may restrict their use for energy generation. Therefore, the current research focuses only on *Rīgas Satiksme*—owned territories with favorable conditions—minimal shading and sufficient open space—such as tram depots, end-station loops, and auxiliary trolleybus parks.

The primary obstacle to ESS integration lies in ensuring electrical safety and system protection. Existing traction power supply networks typically use AC Y/Δ transformers followed by 12-pulse rectifiers. In such systems, a short circuit immediately triggers circuit breakers to isolate the fault. However, in the case of a directly connected supercapacitor bank, the stored energy can continue to discharge into the fault path, causing additional damage to the grid's weakest elements or human body in worst case.

As shown in Fig. 11, if an ESS is connected to the OCS, it must include a dedicated disconnection device capable of rapid isolation under fault conditions. The protection mechanism should also incorporate monitoring of the second and third derivatives of current, as current flow from supercapacitors may reach very high magnitudes while remaining within the nominal load range. Similar to the solutions proposed

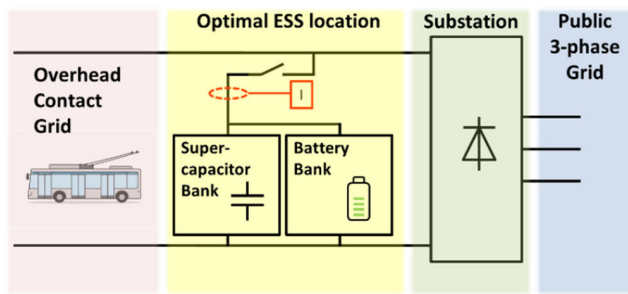


FIGURE 11. Proposed schematic for ESS connection to the overhead contact system (OCS) network. The optimal energy storage system (ESS) location may be at a point along the line (e.g., near a traffic light) or within a traction substation.

in [29], derivative-based fault detection combined with voltage droop analysis offers a more reliable means of identifying short circuits in highly variable load conditions. Another good concept that could be applied in this solution is an “active capacitor” [30] – a large capacitor/super-capacitor equipped by the DC-DC converter that emulates overall capacitor behavior for outer system, while the element itself has a full operation range within safe zone.

It is important to note that the number of vehicles drawing or supplying energy to a given OCS line can vary significantly over time—ranging from multiple simultaneous loads to completely idle states. Therefore, leakage current protection implementation remains a key future research direction. While such protection would substantially enhance human safety, the likelihood of false tripping events during normal operation is expected to be high and must be carefully mitigated through adaptive control and advanced fault discrimination algorithms. Development of protection system is of high importance for future work.

IV. SUBSTATION MODELING IN SIMULINK AND ESS CALCULATION

The physical processes occurring in a trolleybus drive and its interaction with the direct current OCS are highly nonlinear, making their implementation in analytical or coded form extremely challenging. If these processes were purely linear, their modeling would be relatively straightforward. In reality, however, once the line voltage exceeds approximately 780 V, it becomes clamped to prevent overvoltage on the DC line. The DC voltage of the OCS grid is strongly dependent on the instantaneous load conditions and if the other vehicles within the same substation circuit are simultaneously generating regenerative energy. This interaction leads to current circulation phenomenon and system dynamics similar to those observed in weak microgrids.

Such behavior cannot be accurately simulated using simple “hard-coded” equations; therefore, the MATLAB-Simulink environment was chosen for model implementation.

The model shown in Fig. 12 simulates two trolleybuses supplied by the same substation—in this case, it is Substation 17, located in the south-western area of the city. The

model outputs comma-separated value (CSV) files containing voltage and current data for validation, as well as power flow values representing energy consumption and regeneration back to the grid. The independently controlled power chopper begins dissipating excess energy once the DC voltage exceeds 780 V.

Several additional capacitors and resistors were introduced to the model to eliminate algebraic loops and stabilize the simulation process. The selected values of these auxiliary components are negligible and, therefore, do not influence the overall results.

As illustrated in Fig. 13, the general voltage and current trends obtained from simulation follow the measured data to a large extent. Perfect alignment is not expected, as discussed earlier, because most substations also feed tram lines, and the presence of idle trolleybuses at terminal stops or vehicles not included in the dataset introduces additional capacitance and/or variable load behavior. Consequently, measured and simulated current values do not fully match, as it was expected prior. Nevertheless, the overall waveform similarity confirms that the model correctly reproduces the overall system’s dynamic characteristics.

Although the instantaneous voltage levels are slightly lower than the measured ones, resulting in slightly higher current values than expected, the modeled power values—both consumed and regenerated—show only minor deviations from the experimental data.

As it is seen from Fig. 13 and Fig. 14, even a 7 F super-capacitor solution can significantly decrease voltage sweeping and enhance recuperation energy utilization, by almost eliminating rising of the voltage to the level when dissipation resistors are turned on.

To estimate the required battery capacity, the total energy consumption (including recuperation) was summarized for each substation. Calculations were performed for January 6, 2025, which is the date when trolleybus data was received also represents a typical day, and for May 28, representing a day with maximum solar yield according to the available dataset.

As an example, assuming a 200 kW photovoltaic (PV) generation plant, the 90th percentile (p90) solar yield on January 6 covers approximately 3% of the total trolleybus energy consumption. On May 28, the p90 yield would cover around 105% of the total demand for Substation 12, serving as a representative case.

Table 1 provides a comparison of different manufacturer supercapacitors available on the market. The target is to withstand 1200V of DC voltage and series-parallel must be 7 F. To calculate total costs and pay-off year count, additional 30% was assumed for infrastructure-related costs. Bare onboard capacitor solutions were not assigned additional costs, as these units are intended to be installed in cabinets. The stated prices refer to single-unit purchases without price discounts for quantity.

The example observed in Fig. 14 makes modelled capacitor current that can be observed in Fig. 15.

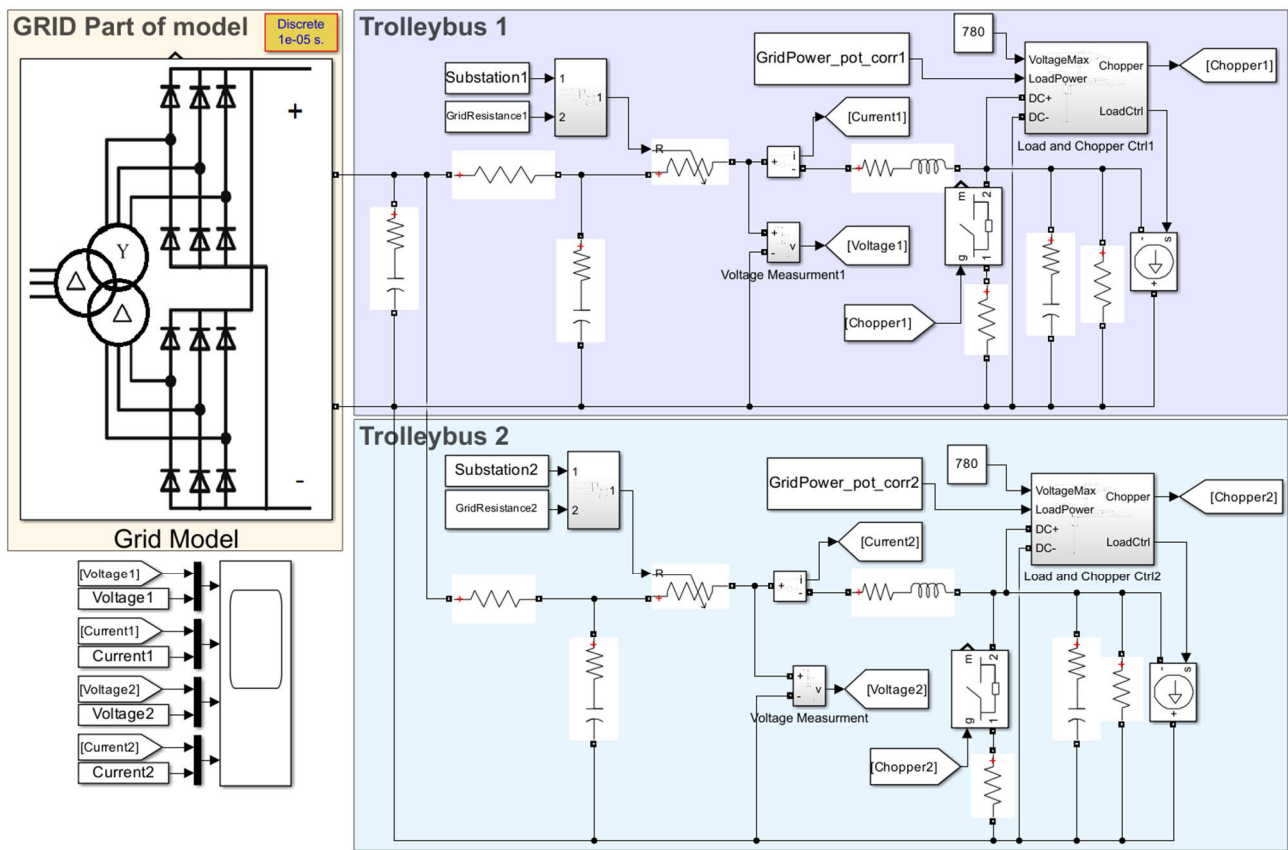


FIGURE 12. MATLAB simulink model representing two trolleybuses connected to a single substation.

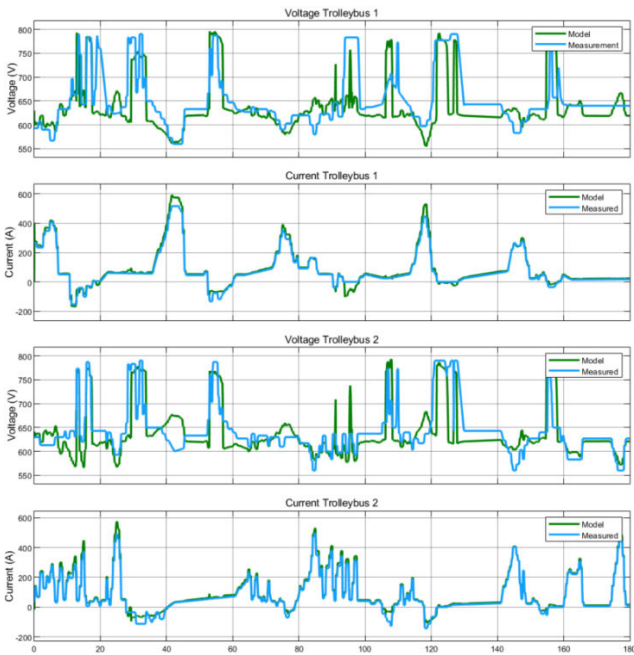


FIGURE 13. Simulation results for two trolleybuses operating within Substation 17, recorded over 320 s beginning at 11:38:07 on 06 January 2025.

TABLE 1. Estimated battery capacities for different operational scenarios.

Model, Manufacturer	Count	Total Price (EUR)	Pay-off years
XVM-259R2635-R, Eaton Electronics	17	63935	3.089
BMOD0006 E160 C02, Maxwell Technologies	80	79768	3.854
LHVS1440625C0B00, Eaton Electronics	18	100620	4.861
XLHVS1440625C0B00, Eaton Electronics	18	100620	4.861
SKELMOD 51V, Skeleton Technologies	24	135626	6.552

As follows from Fig. 14 and Fig. 15, the presented model and calculations indicate that the voltage and current stress applied to super-capacitor are negligible comparing to aging, even when the selected supercapacitor is not a high-performance unit in terms of ESR. High demands for safety set low operation voltage level compared to maximum withstand voltage. Therefore, the expected lifetime tends toward 20 years of use.

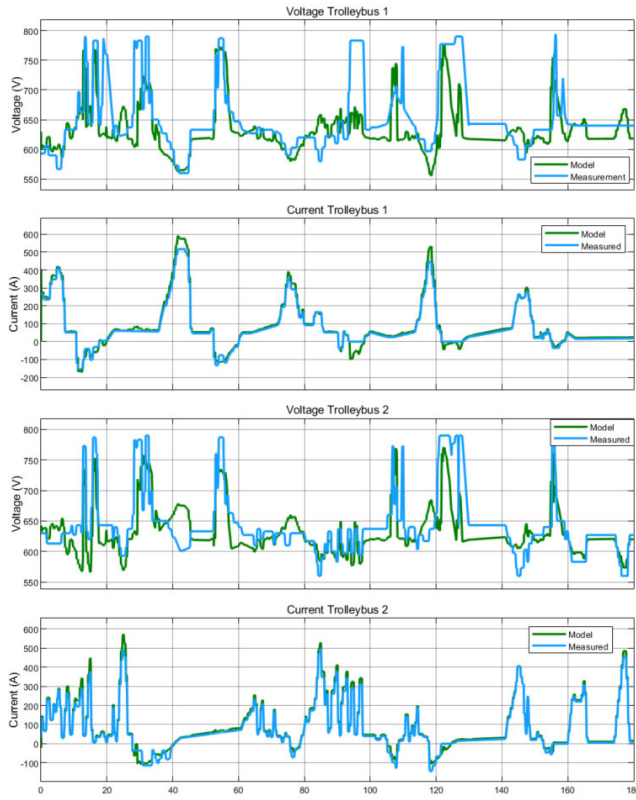


FIGURE 14. Simulation results for two trolleybuses within a substation including a 7 F supercapacitor.

If a higher-capacity solution, such as a Li-ion or Li-polymer battery system, is implemented, additional power-electronic interfaces are required, including a Battery Management System (BMS) and a bidirectional DC-DC converter to utilize the full voltage range of the battery. The optimal charging capacity must be selected according the operational scenario and the system objectives. Five different scenarios can be outlined:

1. Self-consumption (Zero export):
 - a. Goal: To avoid feeding power back into the external grid or keep exports below a defined threshold.
 - b. Rule: When solar generation exceeds local demand, the battery is charged until it reaches its rated power or capacity limit; only surplus energy beyond this point is exported.
2. 24-hour self-sufficiency (Daily islanding):
 - a. Goal: To operate independently on the external grid within each 24-hour cycle if possible.
 - b. Rule: Excess solar energy is stored during daylight hours and discharged during demand peaks, minimizing or eliminating grid imports over the day.
3. Peak shaving:
 - a. Goal: To maintain grid import below a contractual or technical limit.

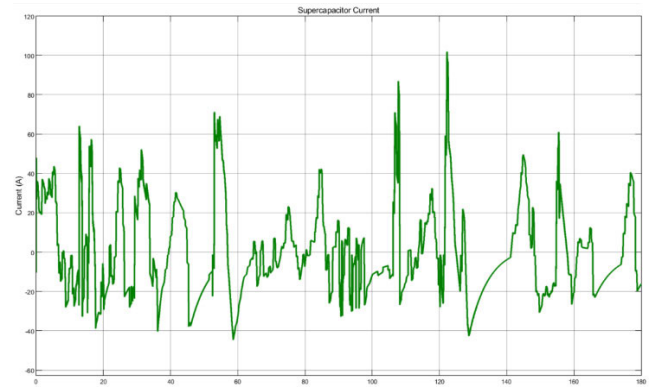


FIGURE 15. Simulated supercapacitor current for a substation supplying two trolleybuses (7 F supercapacitor).

- b. Rule: When demand exceeds the limit, the battery discharges to supply the difference; when load is low or PV generation is available, it recharges.
4. Smoothing (ramping/variability control):
 - a. Goal: To stabilize power exchange with the grid by limiting rapid fluctuations and ramp rates.
 - b. Rule: The battery dynamically absorbs short-term surges and supplies power during dips, maintaining grid power within a defined magnitude and time window.
5. Time-of-Use (TOU) arbitrage:
 - a. Goal: To minimize the total electricity cost according to a time-dependent price tariff (in this study, the Nord Pool price dataset for 06 January 2025 was used).
 - b. Rule: Charge during low-price or PV-surplus hours and discharge during high-price periods to reduce costly grid imports.

TABLE 2. Estimated battery capacities for different operational scenarios.

Scenario	Battery Capacity (Wh)	Payoff Years
1. Zero export	689520	22
2. Daily self sufficient	662152	22
3. Peak shave 150000W	72173.35	11
4. Smoothing 50000W band	435105.5	12
5. TOU 10000Wh (best cost)	9426685	10

Table 2 summarizes the estimated battery capacities (in Wh) required for the different operational strategies. To approximate realistic performance, a charging efficiency of 97%, discharge efficiency of 94%, and an overall converter efficiency of 95% were assumed. PV generation data were taken as average values of two representative dates

— 05.01.2025 and 28.05.2025 (p90) — representing winter and peak solar conditions, respectively. Each strategy was evaluated at 50 equally spaced steps between 10 000 Wh and 100 000 000 Wh. The minimum required battery capacities derived from these simulations are presented in Table 2. Further application of the techniques similar to those presented in [17] could provide more optimal results.

For a preliminary economic assessment, a battery cost of 300 EUR/kWh was assumed. The operating range was limited to a minimum state of charge (SoC) of 0.2 and a maximum SoC of 0.8, while accounting for battery replacement over the system lifetime.

As shown in Table 2, the zero-export and daily self-sufficiency strategies require substantially higher storage capacities, resulting in poor economic performance. In contrast, peak shaving and smoothing strategies demonstrate more practical feasibility.

The time-of-use (TOU) strategy appears economically attractive due to effective utilization of price variations; however, it requires significantly larger storage capacity and more intensive cycling, resulting in accelerated degradation. This raises concerns regarding both economic and environmental feasibility when more detailed analyses are applied.

It should be noted that these results are based on simplified assumptions and should be interpreted as approximate. Further optimization, potentially through hybrid systems combining batteries and supercapacitors, may yield improved performance, however, this lies beyond the scope of the present study.

V. CONCLUSION

Modern big data techniques can provide valuable insights for analyzing complex systems such as electric public transport networks. Studying one or two individual vehicles cannot accurately represent the overall system behavior, as public transport operation follows statistical patterns rather than deterministic rules. Therefore, processing and analyzing large-scale datasets enables a more comprehensive understanding of correlations and tendencies, helps identify optimal locations for renewable energy and energy-saving solutions, and produces datasets suitable for use in simulation environments.

One of the most challenging aspects of data mining in this context is the correct interpretation, representation, and imputation of missing data. The researchers' main task is to maintain the dataset to real-world conditions as close as possible, despite inevitable losses and distortions.

Although the available data are incomplete, it is still possible to determine recommended locations and sizing for ESS and to visualize them clearly. Map-based representation supports better understanding of which urban areas are suitable for PV solar installations and helps to identify new opportunities to enhance infrastructure. Integrating PV systems effectively into the transport grid can bring additional benefits to the public transport fleet, improving both efficiency and sustainability. The research shows that supercapacitors are

a more suitable solution compared to Li-Ion batteries for these applications, or maybe its combination will make it an even better solution. However, some more advanced and less aggressive battery management algorithms would show better profitability.

The introduction of ESS also presents new technical and safety challenges. Traditional OCSs typically designed to maintain structural simplicity for reliability and protection purposes. Integrating large-scale energy storage requires the development of advanced protection and monitoring systems that could, in the future, enable improved fault detection, faster emergency response, and enhanced diagnostic capabilities.

The created dataset can be exported to MATLAB Simulink for simulation-based testing. This provides a robust mathematical framework for studying various system configurations and control strategies under realistic time-domain conditions. While some deviation from measured data is expected, the simulation results reliably reproduce the main operational tendencies and energy dynamics observed in real operation.

Future work will focus on enhancing the battery capacity calculation algorithms, improving grid and trolleybus models for greater accuracy and stability, and developing more sophisticated PV generation models that incorporate different module configurations and integrate them directly with collected real-world data, take in account self-shadowing for some of the azimuth angles.

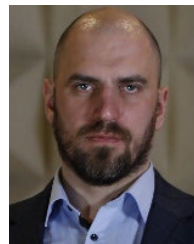
ACKNOWLEDGMENT

Chat GPTo5 model was used to correct the grammar and generate Fig. 10. The underlying idea, data analysis, graphs, and implementation are original.

REFERENCES

- [1] J. Chen, Y. Zhao, M. Wang, K. Wang, Y. Huang, and Z. Xu, "Power sharing and storage-based regenerative braking energy utilization for sectioning post in electrified railways," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 2, pp. 2677–2688, Jun. 2024, doi: [10.1109/TTE.2023.3295089](https://doi.org/10.1109/TTE.2023.3295089).
- [2] M. Khodaparastan, A. A. Mohamed, and W. Brandauer, "Recuperation of regenerative braking energy in electric rail transit systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 8, pp. 2831–2847, Aug. 2019, doi: [10.1109/TITS.2018.2886809](https://doi.org/10.1109/TITS.2018.2886809).
- [3] A. Rufer, D. Hotellier, and P. Barrade, "A supercapacitor-based energy storage substation for voltage compensation in weak transportation networks," *IEEE Trans. Power Del.*, vol. 19, no. 2, pp. 629–636, Apr. 2004, doi: [10.1109/TPWRD.2004.824408](https://doi.org/10.1109/TPWRD.2004.824408).
- [4] H. Zhang, W. Xiang, W. Lin, and J. Wen, "Grid forming converters in renewable energy sources dominated power grid: Control strategy, stability, application, and challenges," *J. Modern Power Syst. Clean Energy*, vol. 9, no. 6, pp. 1239–1256, Nov. 2021, doi: [10.35833/MPCE.2021.000257](https://doi.org/10.35833/MPCE.2021.000257).
- [5] M. Bartłomiejczyk and S. Mirchevski, "Reducing of energy consumption in public transport—Results of experimental exploitation of super capacitor energy bank in Gdynia trolleybus system," in *Proc. 16th Int. Power Electron. Motion Control Conf. Expo.*, Sep. 2014, pp. 94–101, doi: [10.1109/EPEPMC.2014.6980616](https://doi.org/10.1109/EPEPMC.2014.6980616).
- [6] N. Kano, Z. Tian, N. Chinomi, X. Wei, and S. Hillmansen, "Renewable sources and energy storage optimization to minimize the global costs of railways," *IEEE Trans. Veh. Technol.*, vol. 74, no. 5, pp. 7049–7060, May 2025, doi: [10.1109/TVT.2023.3265944](https://doi.org/10.1109/TVT.2023.3265944).

- [7] U. Sirmelis, L. Grigans, and L. Latkovskis, "An analytic simulation model for a supercapacitor-based energy storage system," in *Proc. 14th Eur. Conf. Power Electron. Appl.*, Aug. 2011, pp. 1–10.
- [8] M. Bartłomiejczyk, "Modern technologies in energy demand reducing of public transport—Practical applications," in *Proc. Zooming Innov. Consum. Electron. Int. Conf. (ZINC)*, May 2017, pp. 64–69, doi: [10.1109/ZINC.2017.7968665](https://doi.org/10.1109/ZINC.2017.7968665).
- [9] M. Bartłomiejczyk, "Super capacitor energy bank MEDCOM UCER-01 in gdynia trolleybus system," in *Proc. IECON - 42nd Annu. Conf. IEEE Ind. Electron. Soc.*, Oct. 2016, pp. 2124–2128, doi: [10.1109/IECON.2016.7793086](https://doi.org/10.1109/IECON.2016.7793086).
- [10] T. Zhang, R. Zhao, E. E. F. Ballantyne, and D. A. Stone, "Increasing urban tram system efficiency, with battery storage and electric vehicle charging," *Transp. Res. D, Transp. Environ.*, vol. 80, Mar. 2020, Art. no. 102254, doi: [10.1016/j.trd.2020.102254](https://doi.org/10.1016/j.trd.2020.102254).
- [11] I. Diab, A. Saffirio, G. R. Chandra-Mouli, and P. Bauer, "A simple method for sizing and estimating the performance of PV systems in trolleybus grids," *J. Cleaner Prod.*, vol. 384, Jan. 2023, Art. no. 135623, doi: [10.1016/j.jclepro.2022.135623](https://doi.org/10.1016/j.jclepro.2022.135623).
- [12] S. Kratz, B. Krueger, R. Wegener, and S. Soter, "Integration of photovoltaics into a smart trolley system based on SiC-technology," in *Proc. IEEE 7th Int. Conf. Power Energy (PECon)*, Dec. 2018, pp. 168–173.
- [13] Z. Li, K. Miao, W. Bao, C. Lu, X. Li, and L. Zhang, "Photovoltaic utilization benefit evaluation of flexible DC traction power systems," in *Proc. IEEE 8th Conf. Energy Internet Energy Syst. Integr.*, Nov. 2024, pp. 2899–2904, doi: [10.1109/EI264398.2024.10991505](https://doi.org/10.1109/EI264398.2024.10991505).
- [14] W. Deng, C. Dai, W. Chen, and S. Gao, "Experimental investigation and adaptability analysis of hybrid traction power supply system integrated with photovoltaic sources in AC-fed railways," *IEEE Trans. Transport. Electrification*, vol. 7, no. 3, pp. 1750–1764, Sep. 2021, doi: [10.1109/TTE.2021.3053053](https://doi.org/10.1109/TTE.2021.3053053).
- [15] N. Chinomi, Z. Tian, N. Yang, N. Kano, and L. Jiang, "Analysis of energy efficiency and resilience for AC railways with solar PV and energy storage systems," *IEEE Trans. Ind. Cyber-Phys. Syst.*, vol. 2, pp. 519–530, 2024, doi: [10.1109/TICPS.2024.3468229](https://doi.org/10.1109/TICPS.2024.3468229).
- [16] J. A. Aguado, A. J. S. Racero, and S. De La Torre, "Optimal operation of electric railways with renewable energy and electric storage systems," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 993–1001, Mar. 2018, doi: [10.1109/TSG.2016.2574200](https://doi.org/10.1109/TSG.2016.2574200).
- [17] M. Li, L. Wang, Y. Wang, and Z. Chen, "Sizing optimization and energy management strategy for hybrid energy storage system using multiobjective optimization and random forests," *IEEE Trans. Power Electron.*, vol. 36, no. 10, pp. 11421–11430, Oct. 2021, doi: [10.1109/TPEL.2021.3070393](https://doi.org/10.1109/TPEL.2021.3070393).
- [18] A. Sauhats, D. Zalostiba, A. Utans, and R. Petrichenko, "Power systems transition using biofuels, carbon capture and synthetic methane storage," in *Proc. E3S Web Conf.*, vol. 572, Sep. 2024, p. 2001, doi: [10.1051/E3SCONF/202457202001](https://doi.org/10.1051/E3SCONF/202457202001).
- [19] A. Sauhats, K. Burcevs, R. Petrichenko, G. Bockarjova, and D. Zalostiba, "Minimizing transition costs: Choice of green energy mixes and storage solutions in the Baltic power system conditions," in *Proc. IEEE 65th Int. Scientific Conf. Power Electr. Eng. Riga Tech. Univ. (RTUCON)*, Oct. 2024, pp. 1–6, doi: [10.1109/RTUCON62997.2024.10830817](https://doi.org/10.1109/RTUCON62997.2024.10830817).
- [20] J. Zhang, Z. Tian, W. Liu, L. Jiang, J. Zeng, H. Qi, and Y. Yang, "Regenerative braking energy utilization analysis in AC/DC railway power supply system with energy feedback systems," *IEEE Trans. Transport. Electrification*, vol. 10, no. 1, pp. 239–251, Mar. 2024, doi: [10.1109/TTE.2023.3253860](https://doi.org/10.1109/TTE.2023.3253860).
- [21] D. Iannuzzi and P. Tricoli, "Speed-based state-of-charge tracking control for metro trains with onboard supercapacitors," *IEEE Trans. Power Electron.*, vol. 27, no. 4, pp. 2129–2140, Apr. 2012, doi: [10.1109/TPEL.2011.2167633](https://doi.org/10.1109/TPEL.2011.2167633).
- [22] Z. Gao, J. Fang, Y. Zhang, and D. Sun, "Control strategy for wayside supercapacitor energy storage system in railway transit network," *J. Modern Power Syst. Clean Energy*, vol. 2, no. 2, pp. 181–190, Jun. 2014, doi: [10.1007/s40565-014-0060-4](https://doi.org/10.1007/s40565-014-0060-4).
- [23] R. Barbone, R. Mandrioli, M. Riccio, and G. Grandi, "Development of a high-precision and flexible model for accurate simulation of trolleybus grids," *IEEE Access*, vol. 11, pp. 35022–35034, 2023, doi: [10.1109/ACCESS.2023.3265379](https://doi.org/10.1109/ACCESS.2023.3265379).
- [24] W. Kampeerawat and T. Koseki, "Integrated design of smart train scheduling, use of onboard energy storage, and traction power management for energy-saving urban railway operation," *IEEE J. Ind. Appl.*, vol. 8, no. 6, pp. 893–903, 2019, doi: [10.1541/ieejia.8.893](https://doi.org/10.1541/ieejia.8.893).
- [25] C. Wu, S. Lu, Z. Tian, F. Xue, and L. Jiang, "Energy-efficient train control with onboard energy storage systems considering stochastic regenerative braking energy," *IEEE Trans. Transport. Electrification*, vol. 11, no. 1, pp. 257–274, Feb. 2025, doi: [10.1109/TTE.2024.3389960](https://doi.org/10.1109/TTE.2024.3389960).
- [26] C. Wu, S. Lu, F. Xue, L. Jiang, and M. Chen, "Optimal sizing of onboard energy storage devices for electrified railway systems," *IEEE Trans. Transport. Electrification*, vol. 6, no. 3, pp. 1301–1311, Sep. 2020, doi: [10.1109/TTE.2020.2996362](https://doi.org/10.1109/TTE.2020.2996362).
- [27] M. Khodaparastan, O. Dutta, M. Saleh, and A. A. Mohamed, "Modeling and simulation of DC electric rail transit systems with wayside energy storage," *IEEE Trans. Veh. Technol.*, vol. 68, no. 3, pp. 2218–2228, Mar. 2019, doi: [10.1109/TVT.2019.2895026](https://doi.org/10.1109/TVT.2019.2895026).
- [28] C. Dos Santos Castilho, J. P. N. Torres, C. A. F. Fernandes, and R. A. M. Lameirinhas, "Study on the implementation of a solar photovoltaic system with self-consumption in an educational building," *Energies*, vol. 14, no. 8, p. 2214, Apr. 2021, doi: [10.3390/en14082214](https://doi.org/10.3390/en14082214).
- [29] H. Kumar, S. K. Yadav, K. Sahay, and S. S. Kumar, "Investigation on recuperation of regenerative braking energy using ESS in (Urban) rail transit system," in *Proc. Int. Conf. Electr., Electron. Comput. Eng. (UPCON)*, Nov. 2019, pp. 1–6, doi: [10.1109/UPCON47278.2019.8980241](https://doi.org/10.1109/UPCON47278.2019.8980241).
- [30] H. Wang and H. Wang, "A two-terminal active capacitor," *IEEE Trans. Power Electron.*, vol. 32, no. 8, pp. 5893–5896, Aug. 2017, doi: [10.1109/TPEL.2017.2668764](https://doi.org/10.1109/TPEL.2017.2668764).



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